

Project: Hotel Booking Interface

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Context and Background

Dear Team,

I am excited to share that our small software company has been commissioned to develop a booking app for the Boutique Hotel Technikum. This app will be available on both smartphones and desktops and will be able to handle up to 100 visitors per day. Our goal is to create a modern and sustainable app that can be extended with additional features in the years to come.

The booking app will provide guests of the Boutique Hotel Technikum with a user-friendly way to book hotel rooms for a specific period. Guests will also be able to register in order to conveniently view their bookings and personal details. The project will be carried out in teams of three, with each team responsible for both the backend and the frontend.

We know this project presents a challenge — but we are confident that with our dedication and creativity, we will rise to meet it. We also recognize that we are not only building a great app, but gaining valuable experience and skills as developers.

I look forward to working with you and to sharing our skills and ideas to achieve the best possible result. We will support the project through feedback rounds and code reviews and will assist you whenever needed. Let us work together on this project and create a great booking app for the Boutique Hotel Technikum!

Best regards,
your project lead

Technical Requirements

- The booking app will be developed using Vue.js (Version 3) and implemented efficiently using the Ionic framework.
- The Axios library is used for API calls, and state management is handled using Pinia.
- The application is primarily developed for mobile devices but must also be usable on desktops.
- For the backend we will use Spring Boot (Version ≥ 3).
- A MySQL database is used as the data store.

Best Practices

The following engineering standards and best practices are mandatory throughout the project:

- **Clean Code**
- **Clean Architecture** as defined by R. Martin and discussed in class
- **Richardson Maturity Model** meet Level 2
- **Atomic Design**: the frontend must be structured accordingly
- **Pinia** must be used as discussed in class
- All further best practices discussed and introduced during the course are equally binding and must be applied consistently.

User Stories

U1: Hotel Website

As a guest I want to be presented with the hotel in the form of a website in order to learn more about it.

Details

In addition to the booking and management features, the Single Page Application also contains static pages that invite visitors to explore the hotel and provide them with the information they need to know what to expect before, during, and after a successful booking.

Recommended Approach

- Define what information needs to be provided for the hotel presentation
- Prepare content, research and provide images
- Create a component for each page

Definition of Done

- ☐ HTML templates for the static pages have been created (at minimum: landing page, imprint, about)
- ☐ Texts and content have been prepared and integrated
- ☐ The individual pages can be accessed and displayed correctly in the browser
- ☐ The content is well arranged and readable on both a smartphone and a desktop device

U2: Hotel Room Selection

As a guest I want to see an overview of hotel rooms and their details in order to choose a room that suits my preferences.

Details

An API endpoint is available to retrieve room information. Details can be found in the API documentation.

Room images are located in the `images/rooms` folder of the repository. File names correspond to the respective room ID. The images may not all be the same size — please find a suitable solution for this.

Meaningful icons should be provided for the extras so that the user can navigate easily. A maximum of 5 hotel rooms should be displayed on the first load.

This means a solution is required for how to display an odd number of rooms attractively, and how to navigate to page 2 (and potentially pages 3 and 4 — a room extension is planned for the coming year).

Recommended Approach

- Familiarize yourself with the API endpoint
- Analyze the response data
- Create a paper prototype for the display and pagination
- What interactions are required?
- Familiarize yourself with [Bootstrap Icons](#)
- If you use a different design system, explore the icons available there
- Select icons for the extras
- Use button groups for the pagination buttons

Definition of Done

- ☐ Hotel rooms can be displayed with image, title, description, and extras
- ☐ Five hotel rooms are shown on page one
- ☐ It is possible to navigate to further pages: pagination
- ☐ The extras are accompanied by icons that represent them
- ☐ The feature has been checked into its own branch and pushed to the Git repository
- ☐ The feature has been merged into master after a code review

U3: Check Availability

As a guest I want to check whether a specific room is available for my desired travel period in order to determine whether I will be able to book the room.

Details

An API endpoint is available to retrieve room availability. Details can be found in the API documentation.

The user should be able to define their booking period within the interface. You have full creative freedom in deciding how best to present this to the user.

Think innovatively and outside the box.

Recommended Approach

- Familiarize yourself with the API endpoint
- Analyze the response data
- Develop two to three variants for how the user could define the period, and how room availability could be displayed
- Think through the user flow and use case scenario — what error cases exist?
- Consider which new components need to be designed, or which existing ones need to be extended or modified, so that your GUI concept is also feasible from an implementation perspective
- Build the store and API integration first, then the GUI

Definition of Done

- ☐ The user can define the booking period using a date dialog
- ☐ The user receives clear feedback about availability within the selected period
- ☐ Possible error cases are handled and the user is informed accordingly
- ☐ The feature has been checked into its own branch and pushed to the Git repository
- ☐ The feature has been merged into master after a code review

U4: Book a Hotel Room

As a guest I want to book a selected room for a selected period in order to have accommodation during my holiday.

Details

An API endpoint is available to create a booking for a room. Details can be found in the API documentation.

Once the user has found an available room, they should be able to book it. The following data must be provided:

- First name
- Last name
- Valid email address
- Confirm email address
- Breakfast yes / no

Before final submission, the user should have the opportunity to review their booking details (including the booking period and room selection) and make changes if necessary. After a successful booking, a confirmation must be shown to the user.

Recommended Approach

- Familiarize yourself with the API endpoint
- Analyze the data to be transmitted
- Create a paper prototype including user flow and error handling
- Build the store and API integration first, then the GUI

Definition of Done

- ☐ The user can book a desired room for a desired period by providing complete and correct data, provided the room is available
- ☐ Possible error cases are handled and the user is informed accordingly
- ☐ The user can review their data before submitting
- ☐ The user receives a confirmation after a successful booking
- ☐ The feature has been checked into its own branch and pushed to the Git repository
- ☐ The feature has been merged into master after a code review

U5: Improve Booking Confirmation

As a guest I want to be informed after the booking whether it was successful or failed in order to know whether I can complete my process or whether further steps are required.

Details

The booking confirmation introduced in U4 should be extended to provide the user with comprehensive information about all details and to make arrival and check-in easier. The following booking details should be displayed:

- The booking period
- The hotel room including image, title, extras, and description
- The user's submitted personal data
- Directions to the hotel and any additional relevant information (e.g. train connections)
- Contact options

Recommended Approach

- Consider what other information might be necessary beyond what is described here
- Create a paper prototype of the output
- Can existing components be reused?
- Can Google Maps or a similar service be integrated for directions? Are there packages available, and what is required?

Important Comment

This user story marks (when U1–U4 are also completed) the first milestone and therefore the first release. This brings us to V1.0.0.

Definition of Done

- ☐ The confirmation is extended with the information described in the Details section
- ☐ The confirmation should be printable on A4 as far as possible (printing from the browser is sufficient, no PDF conversion required)
- ☐ There is a clear outcome regarding the directions feature: implementation based on research results
- ☐ The feature has been checked into its own branch and pushed to the Git repository
- ☐ The feature has been merged into master after a code review
- ☐ V1.0.0 has been released (tagging)